

Problem 1

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**Problem 1** (a) Find the linearization,  $L(x)$ , of the function  $f(x) = e^{2x}$  at  $a = 0$ .

(b) Using the linearization,  $L(x)$ , from the part (a), approximate  $e$ .

(c) Is the estimation found in part (b) an overestimate or an underestimate?  
**EXPLAIN.** [2]